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Contention resolution in Optical Burst Switching Networks: a Unified Study of Wavelength Conversion and Optical Buffer Schemes *

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ABSTRACT

Wavelength conversion and optical buffer are the two most commonly used and most useful contention resolution schemes. This paper presents a comprehensive study of them in a multiwavelength optical burst switching network. Specially, we focus on 1) how does the wavelength conversion degree affect the burst blocking performance? 2) how does the optical buffer affect the burst blocking performance? Different from the studies in literature, we will not only investigate the effect of optical buffer length on the blocking performance, but also we investigate the effect of the granularity, which is defined the length of an FDLs and measured in terms of a delay unit, of optical buffers on the burst blocking performance. 3) how does various combinational schemes can be architected and how they perform.

Keywords: Optical burst switching (OBS), contention resolution, wavelength conversion, fiber delay lines, granularity

1. INTRODUCTION

All optical packet switching is quite within the realms of possibility in the future networks. However, the technical difficulties in achieving all optical packet networks lie in the complexity of building large, single-stage all-optical packet switches and lack of scalable optical buffers. In the meantime, optical burst switching (OBS) [1~3] will become a viable intermediate step between current lambda switching and future packet switching as a promising solution for realizing IP over wavelength division multiplexing (WDM).

In an optical burst-switching network, contention occurs at a switching node whenever two or more bursts try to leave the switch fabric on the same output port, on the same wavelength, at the same time. In order to solve this problem, some contention resolution mechanisms, such as wavelength conversion, optical buffer, deflection routing and burst segmentation, have been proposed.

- Wavelength conversion offers effective contention resolution without relying on buffer memory. Wavelength converters can convert wavelengths of bursts that are contending for the same wavelength of the same output port. It is a powerful and the most preferred contention-resolution scheme that does not cause extra latency, jitter, and resequencing problems.
- Optical delay line (which provides sequential buffering) is a close imitation of the RAM in electrical routers,

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although it offers fixed and finite amount of delay. Many previously proposed architectures employ optical delay lines to resolve the contention [7,8]. However, since optical delay lines rely on the propagation delay of optical signal in silica to buffer the packet in time, i.e., due to their sequential access, they have more limitations than the electrical RAM. To implement large buffer capacity, the switch needs to include a large number of delay lines.

- The deflection routing [9~11] approach is a multiple path routing technique. Bursts that lose the contention are routed to nodes other than their preferred next-hop nodes, with the expectation that they will eventually be routed to their destinations. The effectiveness of deflection routing depends heavily on the network topology and the offered traffic pattern. Moreover, deflection could cause bursts to arrive out of order and extra delay.
- Burst segmentation [8,12] is a new contention resolution approach. In burst segmentation, rather than dropping the entire burst during contention, the burst may be broken into multiple segments, and only the overlapping segments are dropped.

Almost all previous studies [4~6] in literature have concentrated on one- or two- dimensional contention resolution schemes of the last three contention resolution schemes and are based on the assumption that every node in an OBS network is with full wavelength conversion capability. However, implementing all-optical full wavelength conversion in an OBS network is quite difficult due to technological limitations and to the fact that to implement such a network is expensive. So, it is interesting to investigate whether we can do as well as full wavelength conversion in terms of blocking performance by using limited-range wavelength conversion. At the same time, the optical buffering approach is the most commonly used and the most useful contention resolution scheme in optical burst switching networks. Therefore, the goal of this paper is to make a unified study of wavelength conversion and optical buffer schemes.

Our focus in this paper is placed on 1) how does the wavelength conversion degree, which will be defined in section 3C of this paper, affect the burst blocking performance? 2) how does the optical buffer affect the burst blocking performance? Different from the studies in literature, we will not only investigate the effect of optical buffer length on the blocking performance, but also we investigate the effect of the granularity, which is defined as the length of an FDLs and measured in terms of a delay unit, of optical buffers on the burst blocking performance. 3) how does various combinational schemes can be architected and how they perform.

One of the important requirements for the simulation experiment is to model the network traffic as close to reality as possible. To realistically simulate the IP traffic, we need to use a proper packet arrival model. It has been reported that the main character of IP traffic is its self-similarity. And it has been shown in literature that self-similar traffic can be generated by multiplexing multiple sources of Pareto-distributed ON/OFF periods [16]. Therefore, we use the Pareto-distributed ON/OFF periods to model IP traffic in our simulation experiment.

The rest of this paper is organized as follows. Section 2 briefly reviews the backgrounds related to this paper. The modelling of IP traffic is presented in section 3. The unified study of wavelength conversion and optical buffer schemes are presented in section 4. And section 5 concludes this paper.

2. BACKGROUNDS

A. Optical burst switching

Optical burst switching provides a mechanism for moving large quantities of bursty data across a transparent optical switching network. OBS can serve as a bridging technology between existing circuit switched transparent optical transport networks and future networks that use pure optical packet switching. In the near term, OBS can be used to

increase the efficiency of existing networks, either by using it in the network core to carry out statistical multiplexing of many streams of variable bit rate data, or by using it at the network edge to aggregate multiple traffic streams onto predefined wavelengths in the core network.

Optical burst switching has been described extensively in the literature; [1] and [2] provide good general discussions of how OBS works. We briefly review the essential details here. Bursts are assembled at the edge of the network by accumulating a set of higher layer protocol data units into a large block of data, called an optical burst. The known burst assembly approaches in literature are timer-based [13] and threshold-based [14]. In the latter approach, once the length of a burst being created reaches a threshold value, the burst is generated and transmitted. In the former approach, the ingress node generates bursts at regular time intervals; the bursts can be of fixed length or variable length depending on whether the burst that is created when the timer expires is padded into a fixed length burst.

When a burst is created, the edge node creates a control packet and transmits it in advance of the burst, usually on a separate wavelength, to reserve resources for the burst at each switch that it will pass through on the way to its destination. The control packet contains all the information required by each switch to forward the burst, such as the length of the burst and the time offset between the control packet's arrival at the switch and the arrival of the burst itself. The time delay between the transmission of the control packet and the launching of its associated burst must be large enough to prevent the burst from overtaking the control packet; the minimum value for the time offset is $H\delta$, where H is the number of hops on the burst's path and δ is the mean time required to process the control packet each switch. Because OBS switches do not convert bursts into electronic form to switch them, the bursts do not experience any queuing delays as they cross the network. The only delays that they encounter are the propagation delays associated with physically traveling along the various fiber links. It is possible for additional delays to occur if multiple bursts arriving at a given switch have to contend for a single output port's resources. Some OBS architectures employ one or two of the contention resolution mechanisms presented in section 1 to resolve this contention.

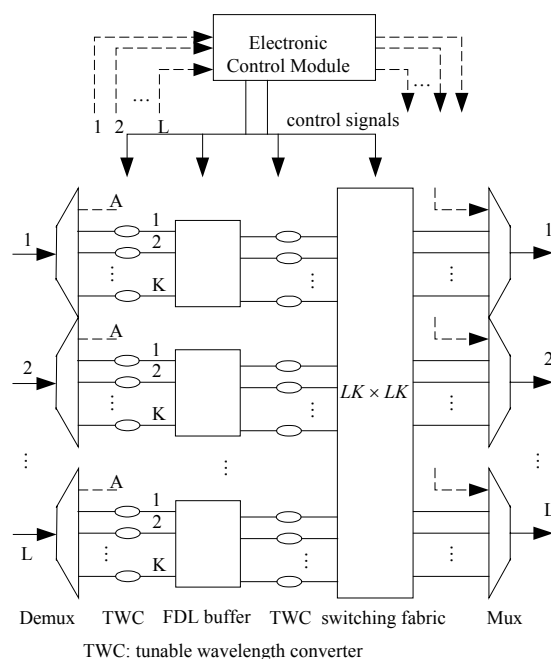


Fig.1. Structure of an optical burst switching node

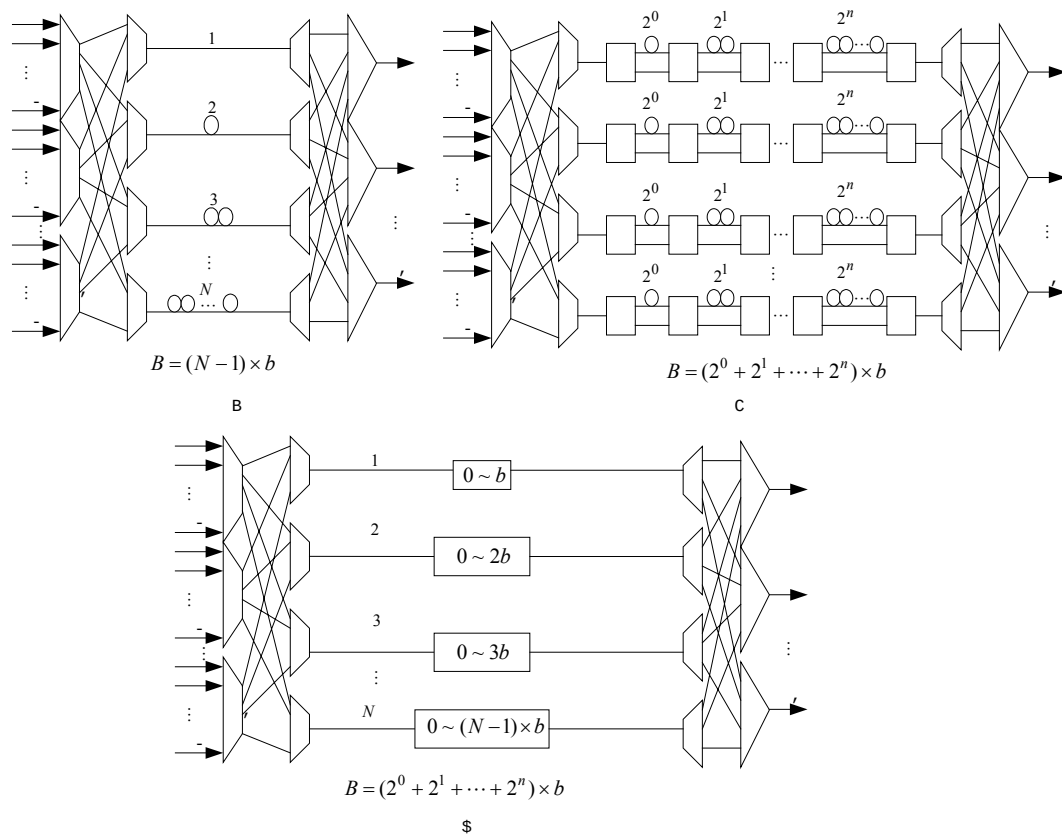


Fig.2. Structure of FDL buffers with (a) fixed delay, (b) variable delay and (c) hybrid

B. Optical burst switching node and optical buffer

The OBS node studied in this paper is first proposed in [4] and is briefly described in this section for complete. The general architecture of an $L \times L$ optical burst-switching node (OSN) is shown in Fig.1. There are L incoming (and outgoing) fiber links, each of which carries k wavelengths for transmitting data bursts (solid lines) and one (or more) additional wavelength for carrying control packets (dotted lines). Every control packet is processed by the electronic control module inside an OSN, which generates appropriate control signals to set up the wavelength converters, FDL buffers, and switching fabric. The optical switching fabric, which functions as an $Lk \times Lk$ non-blocking switch with output buffers, switches each burst on an incoming wavelength as it arrives (i.e., without having to synchronize it with other incoming bursts).

The FDL buffers shown in Fig.1 can be implemented in three ways as shown in Fig.2 (a), (b) and (c). In (a), a FDL buffer has N delay lines, each of which provides a different but fixed delay, which ranges from zero to the maximum buffering time, $B = (N-1) \times b$, where b is the delay time that a single fiber delay element (represented by a circle in Fig.2) can provide. This structure will be referred to as the fixed delay FDL buffer. In Fig.2 (b), a FDL buffer also has the same number of delay lines, N , but each delay line can provide a flexible delay time ranging from zero to $B = (N-1) \times b = (2^0 + 2^1 + 2^2 + \dots + 2^n) \times b$ by using multiple wavelength-sensitive 2×2 switches. The fixed-delay FDL buffer is simpler (i.e., has a lower cost which is proportional to the number of switches), but less effective than the variable delay FDL buffer. The FDL buffer shown in Fig. 2(c) is a hybrid of Fig. 2(a) and (b) where each FDL can provide a

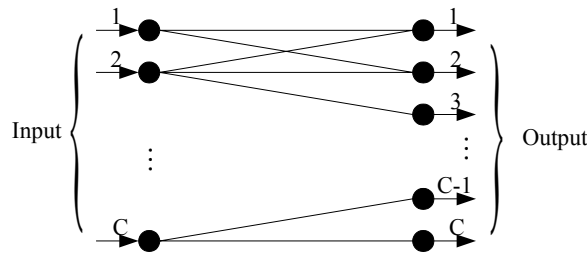


Fig.3. Illustration of limited-range wavelength conversion for $d=2$.

variable delay, but has a different maximum delay ranging from b to $(N-1) \times b$. Note that all the FDL buffers shown in Fig. 2 have $K:1:N$ switches followed by $N:K:1$ multiplexers at the input side, and $N:1:K$ demultiplexers followed by $K:N:1$ switches connected at the output side. This allows one physical FDL to be shared by all K wavelengths, which effectively creates K virtual FDLs per physical FDL. Also, two stages of wavelength converters before and after the FDL buffer as shown in Fig. 1 are useful since they allow a burst to use different wavelengths in a FDL buffer and on an output link, thus resolving contentions among multiple incoming bursts for the same FDL and link more effectively. In the rest of the paper, we will concentrate on the fixed delay FDL buffer and simply refer to it as the FDL buffer, though our discussions also apply to other FDL buffers.

C. Wavelength conversion

There are two types of wavelength conversion. One refers to a limit on the number of nodes with full wavelength conversion capability (also called sparse wavelength conversion). And the other implies that a given input wavelength can be converted into a set of (but not all) output wavelengths, which is also called limited-range wavelength conversion.

In this paper, we only study the limited-range wavelength conversion. Specially, we study the case shown in Fig.3. As shown in Fig.3, for any given wavelength, it can be converted to d adjacent wavelengths on either side of the input wavelength, in addition to the input wavelength itself, where d is the wavelength conversion degree. When $d=n$, the limited-range wavelength conversion becomes the same as full-range wavelength conversion.

In order to compare performance of nodes with different dimensioning, the figure conversion ratio Cr is introduced and defined as the ratio of wavelength conversion degree, d , and the total number of wavelengths, W , i.e. $Cr=d/W$.

3. MODEL OF BURSTY TRAFFIC

It is widely reported that the dominant traffic in future networks is IP traffic. And it is well known that one main character of IP traffic is its burstiness. To include the nature of IP traffic we model it as on-off source traffic with Pareto distribution, which is widely used in literature [15] to simulate IP traffic, as depicted in Fig.1. In the figure, r_{01} denotes the probability of a source going from the idle (off) to the active (on) state while r_{10} denotes the reverse process. Clearly, the mean number of time intervals (time interval = packet period) that the source stays in the on state is given as $1/r_{10} = E[T_{on}]$, where $E[T_{on}]$ is the mean on period. Apparently, the probability for a source to be active is given as

$$P[on] = \frac{r_{01}}{r_{01} + r_{10}} \quad (1)$$

The duration of each on-off period is assumed to be a random variable T_i , $i \in \{on, off\}$, with the Pareto distribution, having a cumulative distribution function

$$F_i(t) = 1 - \left(\frac{t_0}{t_1} \right)^{\alpha_i} \quad (2)$$

with mean $E[T_i] = t_0 \alpha_i / (\alpha_i - 1)$ when $\alpha_i > 1$. In the simulation we set $t_0 = 1$, so that the active period and the idle period of the bursty traffic can vary from 1 to time slots.

The overall network utilization may be written as

$$\rho = E[T_{on}] / (E[T_{on}] + E[T_{off}])$$

Therefore, given the utilization and the mean of the active period $E[T_{on}]$, the parameter α_{off} is obtained as

$$\alpha_{off} = \frac{(1 - \rho) \alpha_{on}}{(1 - \rho) \alpha_{on} - \rho(\alpha_{on} - 1)} \quad (3)$$

Based on (2), given a random variable U with uniform distribution on $[0,1]$, we use the following transformation to generate the random number P of time slots during the active period and the idle period.

$$P = \left\lceil \frac{1}{U^{\alpha}} \right\rceil^+ \quad (4)$$

In order to evaluate the effect of the burstiness of bursty traffic on D , we define the burstiness of bursty traffic as

$$bur = \frac{\text{peak bit rate}}{\text{average bit rrate}} \quad (5)$$

After some simple algebra, we have

$$bur = \frac{1}{r_{10}} = E[T_{on}] \quad (6)$$

4. NUMERICAL RESULTS AND EXPLANATIONS

A. Assumptions

Without loss of generality, the following assumptions are made in this paper.

- We only study a single node case. Therefore, we will treat the notion node and network without difference.
- There is 8 input links and 8 output links at each node and each link contains one fiber.
- There are totally $W+1$ wavelengths in each fiber. Among those wavelengths, one is used to carry the control bursts.
- The wavelength conversion degree is d .
- The data transmission rate is 1Gbps and the switching time is $\delta = 10\mu s$.
- The offset time Δ is calculated out by using $\Delta = H\delta$, where H is the number of hops a burst will encounter from source to destination and is uniform distributed ranges from 1 to 5.
- The first-fit algorithm is used to assign wavelengths for bursts.

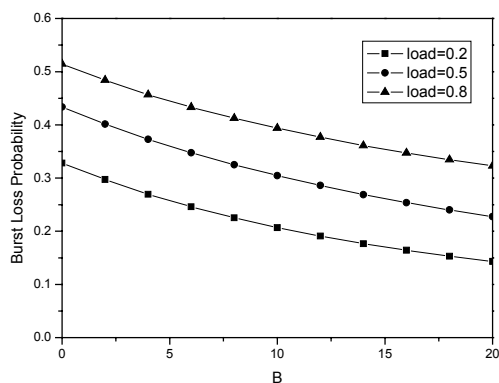


Fig.4 Burst loss probability as a function of buffer length by varying traffic load ρ as a parameter. The granularity in these simulations is 0.06.

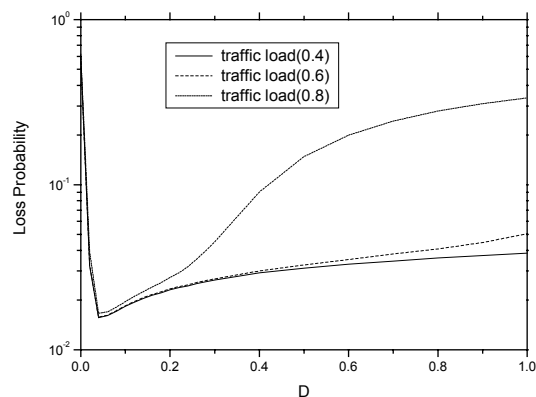


Fig.5 the effect of traffic load on the optimal value of D . Burst loss probability is a function of D varying traffic load ρ as a parameter. The buffer length B in these simulations is 64.

- We use B to denote the buffer length and D to denote the granularity of an optical buffer.
- We use the on-off traffic source with Pareto distribution, which is described in section , to model IP traffic in our simulations.
- The burst arrival process is independent of the state of the queue.
- The lengths of consecutive bursts are also independent and burst lengths are negative exponential distributed with average $L=1/\mu$, where μ is the average service rate, which is independent of the state of the system.

B. Numerical results and explanations

We now study the performance of an optical burst switching node with only optical buffers, with only wavelength conversion and with both optical buffers and wavelength conversion respectively.

1) Optical buffer

The most common way to realize an optical buffer has been by means of fiber delay lines (FDLs). Generally, a FDLs buffer includes two aspects: the buffer length and the granularity, which is defined as the basic time unit of FDLs.

Fig.4 shows the burst loss probability as a function of the buffer length by varying the traffic load. The average burst length is 100,000bits and the granularity is 0.06. Without loss of generality, the value of D is normalized to the average burst length L . From Fig.4, one can see that the burst loss probability decreases with the increase of buffer length. This is because the longer the buffer length is, the more likely is the buffer to delay a burst that lose the contention rather than to discard it. This implies that we can effectively resolve contention by using FDLs.

Fig.5 shows the burst loss probability as a function of the granularity by varying the traffic load. The average burst length is 100,000bits and the buffer length is 64. From Fig.3, we can see that the blocking probability initially reduces with the increasing of D . When it reaches some point D_0 (approximately 0.25 to 0.3), the blocking probability reaches its minimum value. When $D > D_0$, the blocking probability will increases with the increase of D . This implies that there is an optimal value of D , where we can achieve the minimum burst loss probability. Therefore, we should set an optimal value for D when FDLs is used to resolve contentions.

2) Wavelength conversion

Fig.6 shows the burst loss probability as a function of the wavelength conversion ratio by varying the total number of

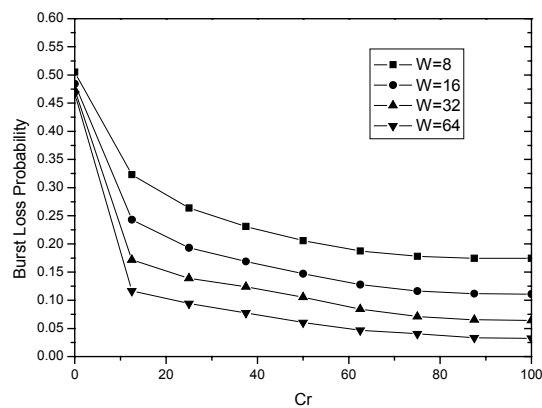


Fig.6 Burst loss probability as a function of D varying the wavelength conversion degree as a parameter. The buffer length B in these simulations is 16 and the traffic load is 0.8.

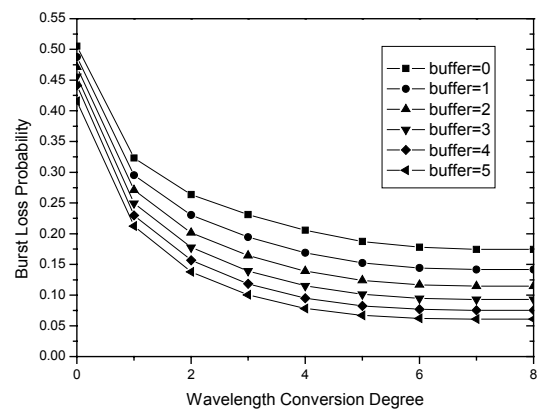


Fig.7 the effect of traffic load on the optimal value of D . Burst loss probability is a function of D varying traffic load ρ as a parameter. The buffer length B in these simulations is 64.

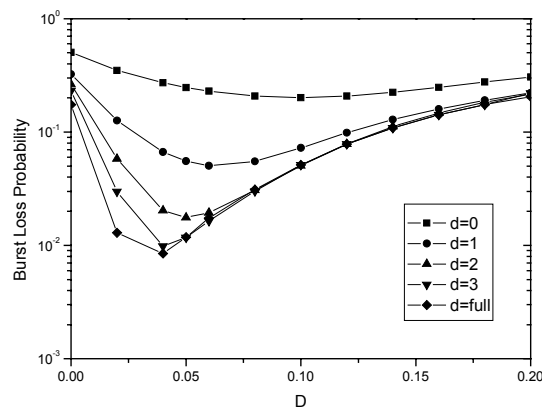


Fig.8 Burst loss probability as a function of D varying the wavelength conversion degree as a parameter. The buffer length B in these simulations is 16 and the traffic load is 0.8.

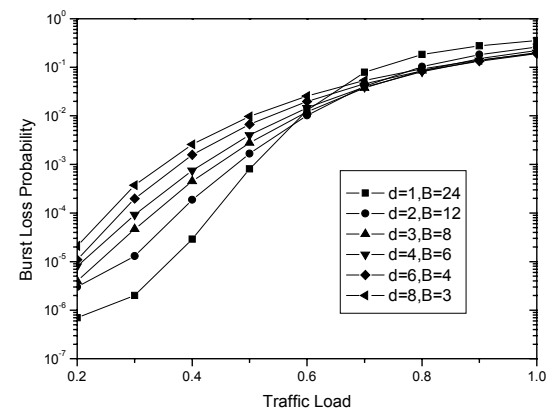


Fig.9 Burst loss probability vs. traffic load. The figure shows different combinations of buffer length and wavelength conversion degree. The product of buffer length and wavelength conversion degree is 24 and the granularity is 0.06.

wavelengths W . One can see that with the increase of wavelength conversion ratio, the burst loss probability reduces rapidly. However, one can also find that a threshold of wavelength conversion (almost 40%) exists for the burst blocking performance improvement. When the wavelength conversion capability exceeds the threshold, the performance improvement is marginal even if more wavelength conversion capability is added.

3) A unified study of wavelength conversion and optical buffers

Fig. 7 shows the burst loss probability as a function of the wavelength conversion degree by varying the buffer length. We simulate eight wavelengths in the network. This is because other number of wavelengths renders similar results, and large number of wavelengths requires much longer time to simulate. The traffic load is 0.8 and the granularity is set to be 0.06 (get from Fig.5). From Fig.7, one can see that, compared to only use one contention resolution scheme, we can achieve better blocking performance by combining the two contention resolution schemes. Fig.7 also shows that wavelength conversion can lead to significant performance improvement in burst loss probability and there exists a

threshold of wavelength conversion degree.

Fig.8 shows the Burst loss probability as a function of D varying the wavelength conversion degree as a parameter. The buffer length B in these simulations is 16 and the traffic load is 0.8. From Fig.8, we also find that there exists an optimal value of the granularity D of FDLs. Further more, we find that the higher the wavelength conversion degree is, the smaller is the optimal value of D . However, we also find that there is a threshold for the wavelength conversion degree d . When the wavelength conversion degree exceeds the threshold, the optimal value of D doesn't change at all. For example, the optimal value of D is approximately 0.04 when wavelength conversion degree is 3. However, the optimal value of D is also 0.04 even the wavelength conversion degree increase to 8, which is the full wavelength conversion case. From the above analysis, we can make the conclusion that we should take care to choose an appropriate value for D in order to efficiently resolve contention if fiber delay lines are used.

From Fig.8, one can also find that wavelength conversion can lead to significant performance improvement in burst loss probability. As shown in Fig.8, the burst loss probability decreases rapidly with the increase of wavelength conversion degree. Also, one can find that there exists a threshold. When the wavelength conversion capability exceeds the threshold, the performance improvement is marginal even if more wavelength conversion capability is added.

Given the product of wavelength conversion degree and buffer length, the various combinations of them are studied in this paper. Fig.9 shows the simulation results. One can see that the best performer under low to medium traffic load (below 0.6) is the combination $d=1, B=24$ and the worst performer is the combination $d=8, B=3$, but the other combinations stand between them. This implies that the optical buffer scheme is more efficient than the wavelength conversion scheme under low traffic load. However, the combinations $d=3, B=8$, $d=4, B=6$, $d=6, B=4$ and $d=8, B=3$ perform as well as each other under high traffic load but the combination $d=1, B=24$ performs the worst. This implies that the wavelength conversion scheme is more efficient than the optical buffer scheme under high traffic load. By comparing the combinations $d=3, B=8$, $d=4, B=6$, $d=6, B=4$ and $d=8, B=3$, we can make the conclusion that a threshold of wavelength conversion exists for the blocking performance improvement.

5. CONCLUSION

In this paper, we make a unified study on two most commonly used and most useful contention resolution schemes, i.e., the wavelength conversion scheme and the optical buffer scheme in optical burst switching networks. We find that optical buffer is an efficient contention resolution scheme. However, there exists an optimal value of the granularity, where we can achieve the minimum burst loss probability.

We find that wavelength conversion can lead to significant performance improvement in burst loss probability. However, it is also found that a threshold of wavelength conversion exists for the performance improvement. When the wavelength conversion capability exceeds the threshold, the performance improvement is marginal even if more wavelength conversion capability is added. We conclude from this that, for an OBS network, limited-range wavelength conversion will be sufficient and full conversion would not be necessary.

We investigated the effects of wavelength conversion degree on the granularity. We found that the higher the wavelength conversion degree is, the smaller is the optimal value of D . However, when the wavelength conversion degree exceeds the threshold, the optimal value of D doesn't change at all. We make the conclusion that we should take care to choose an appropriate value for D in order to efficiently resolve contention if fiber delay lines are used.

By comparing the various combinations of wavelength conversion degree and buffer length, we found that the

wavelength conversion scheme is more efficient than the optical buffer scheme under high traffic load. And the optical buffer scheme is more efficient than the wavelength conversion scheme under low to medium traffic load.

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